

Distributed Systems: Lab 2

Middleware: Object Request Brokers

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1 Introduction

In the previous lab, you implemented a distributed system with a single server and an arbitrary number of clients. The clients were required to explicitly specify the hostname and port of the server in order to interact with the database. It was not convenient as this information could easily change. The situation becomes significantly more complicated when there are many servers joining and leaving the network in an arbitrary way. Such a freedom is at heart of a distributed system, and it is exactly what we aim to achieve by the end of the programming project. In this lab, you will implement the code that will allow the participants of your distributed system to smoothly discover each other. At this point, we shift our focus from the client-server to peer-to-peer model [1] and refer to participants as peers.

The problem outlined above is solved by introducing into our distributed system a so-called middleware [2]; carefully read the corresponding sections in [2] to get a better understanding of the lab. Loosely speaking, a middleware of a distributed system is an auxiliary layer that hides all the complexity of the communication process between peers. Such a layer is typically divided into two parts: object services (e.g., name services) and object request brokers.

A name service is a global registrar of all the objects/peers that participate in the network. Upon request, a name service can, for instance, provide information about the current location of a particular peer; this scenario is depicted in Figure 1. For our purposes, it is sufficient to assume that the name service is unique, it is reliable, and its address is fixed and known to everyone.

An object request broker [2, 3] is a piece of code on a peer's side that interacts with a name service and creates an illusion for the peer that other peers—which might be running on different machines and/or in different processes on the same machine—are running in separate threads of this peer. In other words, it creates an impression that everything is local to the peer, and there is no need of any network communications, which is very convenient.

A name service is already present in the LiU network for you to use. The source code of the name service will not be given; however, the interface of this machine will be described (see Section 2) such that you can employ it for your purposes by sending adequate messages. Consequently, your first task is to learn how to interact with this auxiliary component. To this end, in this lab, you will put aside your previous code and implement a simple application with

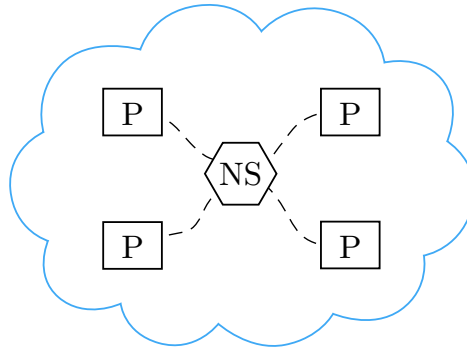


Figure 1: A number of peers (P), equipped with object request brokers, discovering each other through a name service (NS).

the only purpose of registering in the name service and listing the addresses of the peers that are currently present in the network. This knowledge will help you to seamlessly incorporate the name service into your distributed database of fortunes in the future labs.

2 Name Service Interface

The foremost aspect to note is that the name service is following the communication protocol introduced in Lab 1 [4]. Thus, the remote method invocation mechanism of the name service is also based on JSON-encoded messages. The interface has of the following methods accessible to peers:

- **register** — registers a peer in the name service and returns two identifiers assigned to the peer for the future reference (see the next method);
- **unregister** — unregisters a peer in the name service using the two identifiers obtained by calling **register**;
- **require_all** — retrieves the list of the registered peers of a given object type (object types are explained below).

These methods are already being properly called in the provided source code (however, the actual transfer of the associated messages is yet to be fixed by you). Therefore, we leave it to you to find those places in the code and further investigate the input and output arguments of the three methods.

Since there is only one name service for all the students, several students can interfere with each other. In order to prevent this, the name service has been designed in such a way that each student can work within a separate name space. Each name space is characterized by a unique identifier, which we shall refer to as an object type (each peer is an object, and a name space contains all objects of a given type). This unique identifier is to be specified by you in a special file (namely, in `objectType.py` as you will see in Section 3). As the name suggests, you should choose something unique, e.g., your student ID. You can also experiment with other students (but do not forget to tell them beforehand)

by joining several networks together, which can be achieved by sharing the same object type; otherwise, keep your object type secret.

3 Your Task

3.1 Preparation

Continue working with the same code base that you have been working with so far, including all the changes that you have made. The files relevant to this lab are listed below. You should read and understand them.

- `lab2/peer.py` — the `Peer` application (no changes are needed);
- `lab2/test.sh` — a shell script that you can use for testing;
- `modules/Common/nameServiceLocation.py` — the file that contains the address of the name service (no changes are needed);
- `modules/Common/objectType.py` — the file that defines the object type of your peers registered in the name service as explained in Section 2 (should be modified);
- `modules/Common/orb.py` — the code of your object request broker (should be modified);
- `modules/Common/wrap.sh` — the same as for Lab 1 [4].

Go through the code and grasp the main idea behind each part. You can also try to run the main executable file of the lab, that is, `peer.py`, and you will observe that the application cannot even start without your help.

3.2 Understanding the Setup

Let us focus on the module `orb.py` that contains an incomplete implementation of an object request broker [2]. The module includes:

- `Stub` — the class that represents the image of a remote object on the local machine (therefore, it is also called a proxy); the main purpose of the class is remote method invocation [1].
- `Skeleton` — the class that is used to listen to incoming connections and forward them to an instance of the `Peer` class.
- `Peer` — the class that implements basic bidirectional communications using `Stub` and `Skeleton`; any object willing to interact with remote objects should extend this class, which is what is done in `peer.py`.

Now we shall describe how object request brokers work considering a special case wherein a peer interacts with the name service; the same logic applies to communications between two arbitrary peers. Recall that the name service is running on a separate machine in the network. A local representation of the name service is given by the `name_service` instance variable of the `Peer` class. This variable is initialized as an instance of the `Stub` class, which can be viewed

as an image of the name service on the local machine. All this image does (or is supposed to do after you complete the implementation) is the forwarding across the network of incoming method invocations to the name service (the real one).

Stub undertakes the aforementioned forwarding in a very general way, which can be summarized as follows. In **Python**, whenever you call a missing method of an object, an implicit call to the object's `__getattr__` method [5] is done. `__getattr__` exists in all **Python** objects, and it raises an exception by default. In **Stub**, however, this handy method is overwritten in order to generate and return the requested method on the fly. This generated method simply forwards the call to the real object via **Python**'s sockets, which you got familiar with in Lab 1 [4]. The whole idea of **Stub** is to act as an image of any remote object and, thus, to imitate any remote interface. Thus, the three methods listed in Section 2 exist in the name service located somewhere in the network, and **Stub** generates these methods for the client when they are needed.

The **Skeleton** class also plays an important role in our distributed system. In particular, it allows the name service for a proper maintenance of the list of active peers. Whenever a peer leaves the network and unregisters itself, the name service pings the rest of the peers to see whether they are still alive. It is done by calling the `check` method of the **Peer** class, and, if a peer does not reply properly, the name services removes it from the list of active peers. Consequently, without a correct implementation of **Skeleton**, which is supposed to forward incoming requests to **Peer**, all your peers might be removed from the active list of the name service, and a new peer will not see them.

3.3 Implementation

Now you are ready to complete the implementation of the object request broker in `orb.py`, namely, the classes **Stub**, **Skeleton**, and **Peer**. As before, you are supposed to implement the functions marked with “Your code here.”

The requirements are similar to those given in Lab 1 [4]:

- The communication protocol that you should maintain is the one described in Lab 1 [4].
- Your peers should adequately handle exceptions (due to potential network problems, violations of the protocol, invocations of non-existing functions, invocations of proper functions but with wrong arguments, etc.). Please note that you might need to raise a ‘`CommunicationError`’ in certain cases. If you want to be more verbose about the kinds of errors that might, feel free
- Whenever a peer receives an error in response to a method call, it should instantiate and raise this error as if it has occurred locally. Note how you raise exceptions, from the requirements and discussion in Lab 1 [4]. Do not assume that the exception class necessarily exists locally. See lab 1 for the proper behaviour.

As you remember, each remote method invocation should be encoded in JSON and sent to the remote side, which is always the name service in this lab. In this regard, pay your attention to the `_rmi` function in **Stub**. The name service returns the result of the call as a JSON-encoded object containing the output of the method, if it was successful, or an error, if the execution failed.

After a successful completion of the lab, each new peer started in a new terminal window should output its own address and the addresses of all other peers that have already been started and have not been terminated. However, those other peers will not (and, in this lab, are not supposed to) see the new one. Remember that you are provided with a test script (`lab2/test.sh`), which can set up everything for you.

3.4 A note on manual testing

A reminder: testing your implementation manually! That is, try communicating manually with both the stub and the skeleton using “nc”¹. We will not copy-paste the entire section from lab 1 Lab 1 [4], but feel free to open that text.

This note will be relevant to later labs as well, but it will not necessarily be copied into all the text.

3.5 Additional questions

When demonstrating this lab (as with any lab in the course), you should be able to explain your code, but also to evaluate it, and relate it to course concepts. In many cases this will involve discussing design choices and simplifications that have been made for you, and may be somewhat speculative. In some cases you might be asked to answer questions with (at least in retrospect) quite obvious answers. In this lab, this might include questions such as

- How could you (re)implement lab 1 using the middleware from lab 2? If you open the client and server code, which parts would you want to keep?
- Let’s say we have a stub for a remote database object. How much database state is kept in that local object?
- The illusion we want to preserve is that the remote object is a local one, and we are attempting to build a rather generic setup (where you can provide stub/skeleton pairs for many types of objects). Are there any restrictions (assuming that communication works perfectly) on the kinds of objects or return values that we can wrap, while preserving this illusion?
- In this implementation we both have both a Skeleton class and a Request class. Concretely we may have skeleton instance wrapping a database object. What additional functionality is gained by having this separation, rather than just invoking the method on the Skeleton object’s owner directly? Does this introduce any restrictions on the class that we provide a skeleton for?

We do not require written answers to this, and you might not need to provide a solution to all issues raised by the questions.

4 Conclusion

Having completed the lab, you have made an important step towards your distributed database of fortunes: now you have an object request broker that

¹See <https://wizardzines.com/comics/netcat/> or <https://linux.die.net/man/1/nc>

can properly interact with the name service (and, actually, with other peers). However, the simple listing of available peers achieved so far is not something extremely useful by itself. Moreover, as it was discussed at the end of the previous section, each peer is aware only of those participants that joined the system before it. In addition, if a peer leaves the network, the other machines will not notice this change and, thus, can be misled. Therefore, in the next lab, you will (a) extend the application constructed here to call some useful functions of other peers (not only of the name service) and (b) implement an auxiliary class that will help to properly maintain the list of active peers.

References

- [1] <https://www.ida.liu.se/~TDDD25/lectures/lect3.pdf>.
- [2] <https://www.ida.liu.se/~TDDD25/lectures/lect4.pdf>.
- [3] https://en.wikipedia.org/wiki/Object_request_broker.
- [4] <https://gitlab.liu.se/LiU-TDDD25/labs/raw/master/doc/lab1.pdf>.
- [5] <https://docs.python.org/3/reference/datamodel.html>.